Methodology Framework: Dynamic Performance Estimation Model

INPUT

Raw Judge Scores
(S1-S34)

Elimination History
(Sequential Ranks)

Season Metadata
(Judge Counts/Cap)

PROCESS

Adaptive Normalization Operator (Φ_w)
Mapping to [0, 1] Elastic Niche

Pareto Truncated Inversion
$heta_i \propto (1/rank_i)^{\alpha}$

MODEL

Dynamic Performance Evaluation Model
$S_{total} = w_j \cdot s_{judge} + (1 - w_j) \cdot s_{fan}$
MLE Optimization for α^* and w_j^*

EVAL

JPADI Index
Judge-Public Alignment
Drift Measurement

Accuracy Verification
(Predicted vs Actual)
94.7% Success Rate

Uncertainty Quant.
Bootstrap Resampling
Monte Carlo Sim.